

Avery Tanaka

Embedded Systems & IoT Firmware Engineer • 11 Years Experience • MERIQ Index: 74% (Developing)

■ avery.tanaka479@example.com | ■ +1 (351) 711-6184 | ■ Austin, TX

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Python, PCB Debugging, Embedded Linux, ARM Cortex, C/C++, MQTT

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Intel* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Python service, reducing latency by 51%.
- Led a squad of 8 engineers delivering scalable microservices processing 39M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 40%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2015 - 2023)

- Designed and maintained PCB Debugging data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using Embedded Linux saving \$34k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, PCB Debugging, Embedded Linux, ARM Cortex]
Engineered end-to-end production solution leveraging Python and PCB Debugging with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [Embedded Linux, ARM Cortex, C/C++, MQTT]
Architected high-resilience system integrating Embedded Linux with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Cybersecurity and Information Assurance

National University of Singapore — 2011 - 2015 (Score: 3.71 / 4.0)